

INDIAN INSTITUTE OF TECHNOLOGY GANDHINAGAR

EM 630 · Machine Learning and Data Analysis

Course Project Proposal

Day-Ahead Hourly Load Forecasting

for Two 11 kV Feeders in Rajkot City

Time-Series Forecasting · Paschim Gujarat Vij Company Limited (PGVCL)

SUBMITTED BY

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Abstract

This project aims to forecast the next day's hourly electricity demand for two 11 kV feeders located in Rajkot city under Paschim Gujarat Vij Company Limited (PGVCL). Feeder-level load forecasting can help a distribution utility understand expected demand, identify likely peak-load periods, and support short-term operational and maintenance planning.

The project will use three years of synthetic hourly feeder data covering 2023 to 2025. It will be developed as a time-series forecasting project because electricity demand depends on historical consumption and recurring hourly, daily, weekly and seasonal patterns. External factors such as Rajkot temperature, humidity, weekends and Gujarat public holidays will also be considered, as they may influence consumer demand.

The dataset will be checked for missing values, duplicate records and time gaps before studying its major load patterns. Historical load values, calendar information and external factors will then be used to develop a same-hour-previous-week baseline and two main forecasting models. Earlier observations will be used for model development and validation, while 2025 will be kept for final testing. Performance will be compared using suitable regression evaluation measures.

The final system will generate 24 hourly load predictions for each feeder. A simple dashboard will demonstrate the forecasts and compare predicted load with simulated observed load. Synthetic weather and calendar inputs for 2026 will be used to demonstrate future forecasting without depending on a live weather service.

The project focuses on demonstrating a practical and leakage-safe forecasting methodology. Since the feeder readings and future inputs are synthetic, the results will represent a methodological study rather than actual PGVCL operational forecasting accuracy.